Hebbot robotics literature 2026

Finish touch stacking before training reflexes

Hebbot should finish touch stacking first, promote it as a frozen, LLM-free skill, and test a small learned policy only after self-collection has produced about 50 verified successes. The 2026 literature supports that order for three reasons. Contact information decides the last stage of placement on real arms, the credible compression results so far compress agents into code rather than networks, and no paper found trains a neural reflex from anything near Hebbot's 5 verified successes. The strongest counter-evidence is also specific: nobody has published block stacking on Franka joint torque alone, almost every tight-tolerance result uses a wrist force/torque sensor or tactile fingers, and most of the best-fitting papers use cameras Hebbot's assumed wrist-only rig lacks. The project's own record softens the first risk, because the seven defects that stopped touch stacking on 2026-09-01 were geometric bookkeeping rather than sensing limits, and June's tower suggests stacking tolerates several millimetres. The layered thesis has good support at task level and almost no direct test of its central claim, that deliberation compresses into reflex, so a measured practice curve from Hebbot would be close to primary evidence. Nearly all of this rests on unreviewed preprints reporting authors' own trials at 10 to 50 per condition, so the ranking below comes with explicit stop rules.

Most evidence is preprint-grade, and four points in the notes needed correcting

Each section below separates three kinds of statement. **Sourced** claims come from the research notes and carry links. **Project record** claims come from `lab_notebook.md` or the repository, which I checked directly. **Inference** is my reasoning and has no external validation. Unless a venue is named, a cited paper is an arXiv preprint and the numbers are its authors' own. "Not found" means absent from these searches, not proven absent. Search coverage was partial: CoRL 2026 runs this week with 687 listed papers and was not systematically filtered, and ICRA and IROS 2026 proceedings were not swept (alphaXiv). The live Hebbot store was not available, so the 5-in-76 count is taken as given. It has a 95% exact interval of 2.2% to 14.7% (recomputed).

Project record. Two assumptions in the notes do not survive a check against the repository. First, the notes treat 121 probes and 405 s as the current search. Commit `11985a0` already reduced the ±90 mm grid from 121 points to 25, and reaching a 76 mm offset from 26 probes to 2. Notebook #155, written after that commit, still measured about 6.75 minutes per find. The notes' 3.35 s per probe (405/121) is therefore unsafe, and per-probe time has to be measured. Second, the wrist-only assumption may be conservative: the wrist camera is a D405, the

camera code maps a D435 to a `scene` role, and the framing-post draft lists more D435s and two ZEDs as owned. Whether a fixed camera is mounted today is not recorded in what I read.

Two more corrections come from comparing the notes against each other. The contact note leaves the force source of Residual Compliant Policy unresolved, but the distillation note read the full text and found a Resense HEX21 F/T sensor (arXiv 2610.14035). Its 7-demo result is therefore not joint-torque evidence. And the notes disagree on whether simple force-aware VLAs help: TA-VLA, the joint-torque-input model closest to Hebbot's sensing, scores 63.75% against π0.5's 47.5% in FAWAM's table (FAWAM) but 37.5% against 39.0% in Stanford's (FACT). Tasks and evaluators differ, so neither figure should be quoted alone.

Contact decides the last stage, but joint-torque stacking remains unpublished

Sourced. Across the 2026 work, adding force raised contact-rich success substantially, and how force was fused mattered as much as having it.

Work (latest stamp)	Force source	Comparison	Trials	Evidence flag
FR3 vision-torque fusion (Aug)	Franka joint external torque only	Vision-only 30.0%, concatenation 30.0%, gated fusion 82.0%	50 per method	Closest sensor match; no insertion or stacking; 110 to 250 demos; code released
FILIC (Jun)	Wrench estimated from torques, 6-DoF arm	Position-only 26.6 to 46.7%, estimated force 76.6 to 83.3%	30 per condition	Real clearances unstated
FACT (Oct)	Wrist F/T, FR3	π0.5 39.0%, ForceVLA 40.5%, FACT 66.0%	40 per task	Largest trial count; better sensor than Hebbot
FAWAM (Oct)	Wrist F/T, FR3	π0.5 47.5%, FAWAM 85.0%	20 per task, 4 tasks	Eight A100s; no clearances
From Reach to Insert (May)	Tactile fingers, Panda	0.25 mm: 72% to 93%; 0.05 mm: 52% to 67%	75 per clearance	Abstract and intro disagree on the last figure
FORCA (Oct)	Wrist F/T, UR5e, coding agent	Drawer 100% vs 40%; door handle 100% vs 10%	10 per task	Only LLM-commanded force test found

Work (latest stamp)	Force source	Comparison	Trials	Evidence flag
Force Policy (RSS 2026)	Flange F/T, Flexiv	EV plug, final stage: π0.5 0%, Force Policy 65%	10 to 20	Peer reviewed

The most instructive single row is inside Force Policy: RISE-2 scored 100% at contact, 90% at match and 0% at plug-in (RSS 2026). FORCA supplies the matching pattern for an agent, in which generated programs probe a surface, update height or grasp position from pose at contact, and use that value later in the same rollout (FORCA). Joint-torque noise sets the limit. On a Franka, free-space external-torque error is about 0.45 Nm, and FACTR 2 uses the factory signal as its reference, so it cannot show an improvement over it (FACTR 2). A wrench estimate also reads zero near singularities (libfranka). No result at 0.1 mm clearance or tighter used joint torque alone, and no Franka joint-torque block-stacking paper was found.

Bayesian contact search shows how much probing can shrink. A Panda with a wrist F/T sensor reached 6 mm pose error in 8.6 ± 4.5 action cycles using force alone (replayed recordings), and free-space evidence from probes that touched nothing gave the largest single gain (arXiv 2603.21410). BayesContact selects probes by expected entropy on guarded vertical dips and converged to about 4 mm on a KUKA with an ATI sensor (arXiv 2607.16123). The older YCB block protocol records exactly Hebbot's September problem: HERB grasped reliably and failed placement because the block's pose in the gripper was unknown (Calli et al.).

Project record. Finding #156 lists the seven defects that stopped touch stacking, "every one of them in the explicit-geometry pipeline", ending with a 45 mm clustering radius that split a 55 mm block into two objects. June's tower gives a measured tolerance. Finding #135 reports that an 8 mm parallax error stayed "inside the seat margin" and 16 mm did not, with the held block servoed to 1.7 mm over its target. Finding #136 identified `O_F_ext_z` as the right contact signal because a held block slips upward and hides the position error, yet `place_on` in the touch script still descends to a fixed height. The September blocks are 55 × 28 × 27 mm, and June's are described as cubes 2.4 to 2.9 cm tall.

Inference. Stacking is a millimetre-grade task, so the 0.02 to 0.05 mm insertion papers bound what is possible and do not set Hebbot's requirement. The 3 to 6 mm residuals in contact-search papers sit inside the June margin, although those papers used better sensors. Vertical dips only need to detect a stop against a stiff support, which is the regime where joint torque is strongest, and Hebbot's 0.1 mm table probe is z-repeatability, not lateral accuracy. Lateral accuracy comes from the geometry of many dips. One defect looks live in `scripts/stack_by_touch.py`: the fine raster spans ±36 mm around the first coarse hit, not around the object, so a first hit near an edge biases the centroid. By desk arithmetic for an axis-

aligned block with a 20 mm fingertip, worst-case bias is about 6 mm on the short axis and 18 mm on the long axis. Jaw closure should centre the carried block across its short axis, but the base block's 6 mm alone uses most of the June margin. Nothing here has been measured on the arm. A second raster centred on the first centroid, or edge bisection, removes it. Keeping misses as free-space evidence is the cheapest improvement the literature offers. FORCA's pattern maps to probing the base top and the held block's underside relative to the fingers before release.

Targetless calibration now catches gross error but cannot replace touch

Sourced. AnchorHEC (2 Nov) reports 1.22 to 1.35 cm and 1.11 to 1.48° on a physical wrist camera without fiducials, but on an xArm6, and states that calibration after remounting remains untested (AnchorHEC). SyncHeC reports 0.21 cm against a marker board and an 8.1-point policy gain, from a single author, abstract only (SyncHeC). No code release was confirmed for either. No paper was found that detects a loose wrist mount from a residual time series, or that closes the loop from touch points to camera extrinsics. The nearest engineering example is a runtime monitor that treats occluded views as inconclusive (cascade PR).

Inference. Finding #156 says the residual history was the signal and was never examined. That points to a cheap and apparently unpublished instrument: log every (pixel, depth, arm pose, touched point) tuple, run a change-point test on the see-versus-feel residual, and fit disjoint pose subsets so that a sagging mount shows up as residuals correlated with wrist orientation. Finding #155 already gives its value: a better camera model buys search time, not capability. At 1.5° and 300 mm working distance, AnchorHEC-class accuracy is about 8 mm, enough to catch a 14° twist and not enough to place a block.

Evidence quality. Direction of the force effect: moderate, consistent across labs. Size of the effect: low, because baselines, sensors and demo counts differ, and 10 to 20 trials per task leave wide intervals. Transfer to Panda joint torque: low and largely untested. Self-calibration: low, with new preprints and no replication.

Reflexes compile to code long before they compile to networks

Sourced. The cleanest compression numbers are for frozen code. Kuroki et al. (30 Oct) built closed-loop programs from 10 demos per task, replacing about 52.6 model calls per episode with zero, and scored 11/15 on a physical G1 (arXiv 2609.19906). RAX (15 Oct) spent about 75 minutes of tool calls on a first air-fryer run and then executed a compiled program; programs raise `Unsure()` to return control to the agent (RAX). ASPIRE offers the best real-robot token evidence: with a skill, lifting a can took 6.58M tokens to first success rather than 61.94M and rose from 13/20 to 19/20, although skills were learned in simulation and verified only once at

admission (ASPIRE). No real-robot paper was found that plots LLM calls or dollars against practice, and none that distils tens of real LLM-agent episodes into a neural policy.

At small scale, decomposition beats monolithic behaviour cloning in the best-supported comparisons. MT3 (Science Robotics, 2025) outperformed methods given 50 demos using 3 demos per task, by alignment plus open-loop replay (MT3). Tether warped trajectories from 10 demos on a Panda, then autonomously logged 1,085 successes in 1,946 attempts over about 26 hours with 5 interventions, and Diffusion Policies trained on those successes approached policies trained on human demos (Tether). Residual Compliant Policy reached 40/40 against 35/40 for its base policy from 7 demos, using a digital twin and a wrist F/T sensor (RCP). Below 10 demos, neural policies become competitive mainly through heavy augmentation, as in FTE's 256 synthesised demos from one (FTE).

Published autonomous collection runs at roughly 40 to 75 attempts per hour. Tether's labeller, a frontier VLM comparing pre and post images from three views, reached 98.4% precision and 89.6% recall (Tether). AutoDex labels physically, requiring the object to stay at least 5 cm up for a 3 s hold, and reports 5.3% ejection on release, but uses 20 cameras and did not measure label accuracy (AutoDex).

For gating, N-SCORE (RSS 2026) is the strongest available tool: an anytime-valid sequential test whose partial-credit metrics reached decisions in 12 to 36 trials on 4 of 5 hardware tasks, while binary tests failed to decide on 3 of 5 (N-SCORE). A single-author simulation shows McNemar's test with peeking running at about three times its nominal false-positive rate (AgenticRobotics). All these tests assume independent trials. No published demotion procedure was found, and one simulation study warns that a poorly chosen fallback substantially raised cost and collisions (Confidence-Gated Autonomy).

Project record. `src/hebbot/substrate/skill.py` implements `classical` and `composition` bodies and stubs `script`, `trajectory`, `learned_policy` and `vlm_grounded`. `scripts/stack_by_touch.py` runs the arm directly without the MCP server, so its runs do not create substrate episodes.

Inference. "Reflex" should mean any artifact that runs without LLM calls and clears the gate. Under that definition Hebbot can promote a composition body today, whereas a network needs a new body runner first. Trajectory-warping needs a `trajectory` runner, and any of the 5 verified grasps recorded before the September bracket fix have suspect visual alignment even if their arm motion is sound. The notes' yield model of 10 hours to 50 successes borrows Tether's 75 attempts per hour. At 6.75 minutes per find, a two-block attempt spends about 13.5 minutes searching, or about 4 attempts per hour, so 100 attempts at 50% success cost about 22 hours of search alone. Cycle time and success rate are therefore both binding, which raises the value of shrinking the prior. To claim 80% with one-sided 95% confidence takes 14 straight successes; 90% takes 29. Paired trials and stage-level partial credit (base found, grasped,

placed, height verified) are the efficient route, with stages fixed before the run. Bracket and calibration changes break independence, so any mechanical change should expire the skill's evidence.

Evidence quality. Code-as-reflex: moderate in direction, low in sample size, and RAX's real-robot success is judged by the agent itself. Few-demo warping: moderate, but Tether's tolerance figures were not extracted and Tether uses two ZEDs plus a wrist view for labels, so its labeller figure does not transfer to one wrist camera. Sequential testing: high on method, with N-SCORE peer reviewed.

Agents succeed when measurement sits below the LLM

Sourced. Since July, real-robot agent systems have converged on a coding agent over a small primitive set with per-primitive traces and evidence-based promotion. RAX's public schema pairs a `SKILL.md` with `run.py` returning `{ok, measured, why}` and an `evidence/runs.jsonl`, and marks a skill `verified` after at least 2 unattended successes (skills-lib). ARCHITECT on a DROID Panda exposes `move_ee_guarded` and `verify_grasp` as tools. Its mean of 0.81 requires an expert correcting in natural language, falls to 0.38 with VLM corrections, and compares with π0.5 at 0.36 and CaP at 0.18 over 400 rollouts; that mean was computed for the notes rather than reported, and one per-task figure needs checking against the paper (ARCHITECT). SPINE's honesty rule is the most portable idea: a case closes only when live checks pass after the most recent intervention, and memory stores a fix only with a passing record (SPINE). ASPIRE requires a fresh verifier actor to reproduce a success before promotion (ASPIRE).

For memory, PhysMem's simulated ablation favours distilled, verified rules over raw recall: 76% for the full system, 51% with principles only, 28% with episodic memory only and 23% with direct retrieval (PhysMem). Its real-robot evidence is 10 trials per condition, and no adequately powered real-robot test of retrieval against no retrieval was found. RoboVista found that in-context examples lowered accuracy by 2.8 to 6.5 points on its question set (RoboVista).

VLMs remain poor geometric judges. Hebbot's local `qwen2.5vl:7b` scores 43.7% on RoboVista's five-way questions and the best model, Gemini 2.5 Pro, 56.5% (RoboVista). Frontier and robotics-oriented VLMs stayed below 50% macro-F1 on collision grounding in simulation (TouchSafeBench). Abstention is weak by default and improves sharply with defensive prompting, from 16.5% to 93.6% for Gemini Robotics ER 1.6 (RoboAbstention). A dedicated success detector beat VLM-judged termination at 57.39 to 43.50 on long-horizon tasks (Hi-VLA). The apparent contradiction with that paper's finding that VLM size barely matters resolves on role: small models can plan, but they cannot measure.

Robocurve's direct-control benchmark reports top averages of 45 to 48% and $0.38 to $4.51 per subtask under a 40-action cap, but a GPT-6 Astra instance grades the footage without

human validation found (Bitter Lesson paper). A separate safety benchmark concludes that time-critical invariants must belong to deterministic monitors (Bajrami et al.).

Inference. Hebbot's skills-as-records design matches the convergent schema. The two additions worth making are a structured `Unsure` return and expiry of evidence on hardware change, which ASPIRE lists as a limitation and only a simulation-only paper treats. Hebbot's gate should stay stricter than RAX's 2 unattended successes. The local VLM should seed searches (finding #154 puts it within about 20 to 35 mm, inside the coarse step), veto, and never grant a "clear" alone. Reporting false "clear" and false "blocked" separately follows the verifier survey's advice (No Free Checker). SPINE's rule answers the retraction in finding #152: no causal claim enters the notebook until a post-intervention measurement could have falsified it.

Evidence quality. Architecture convergence: moderate, across independent groups. Success rates: low, because trial counts are small and several are self-scored. VLM geometric weakness: moderately high, consistent across benchmarks, mostly on question sets or simulation. Disclosure: this report was written by a Claude model, and Claude-family models appear in several cited systems and as Hebbot's orchestrator.

A 16 GB card fine-tunes only small models, and the smallest VLA trails on Franka

Sourced. No source benchmarks an RTX 5060 Ti, so all figures below come from other GPUs.

Model	Inference	Fine-tuning	Evidence on arms	Fit for Hebbot
ACT	Not stated	2 to 6 GB at BS 8	PhAIL Franka: 10.5% of human throughput, 2.0% interventions; SO-101: stacking 0	Local; first learned test
Diffusion Policy	Not stated	8 to 14 GB at BS 8	Trained on Tether's self-collected data	Local at reduced batch
SmolVLA	Consumer GPUs	10 to 16 GB at BS 8	PhAIL: 6.4%, 18.6% interventions; SO-101 stacking 0.2	Fits; weakest on Franka
π0.5	8.87 GiB peak on a 4090, 3 views	LoRA over 22.5 GB	PhAIL: 13.8%; SO-101 stacking 0.4	Rent training; inference tight
GR00T N1.7	16 GB or more	40 GB or more	N1.6 in PhAIL: 13.3%	No headroom

Model	Inference	Fine-tuning	Evidence on arms	Fit for Hebbot
MolmoAct2-DROID	About 88 GB float32	Not stated	Requires external and wrist cameras	Out of reach

PhAIL, a single-author Franka study with 449 episodes, default recipes and at most 30 trials per cell, found the best VLA about 7× slower than a human (PhAIL). VLA-REPLICA used 50 demos per task on SO-101, where stacking values in 0.2 steps imply about five trials (VLA-REPLICA). The orderings conflict: ACT beats SmolVLA on PhAIL's Franka and loses on VLA-REPLICA's SO-101. Force Policy's authors moved π0.5 from a 24 GB RTX 3090 for memory reasons, and the notes do not settle whether that was training or inference (RSS 2026). Stanford attributes VLA precision failures partly to the flow noise schedule starving low-noise timesteps, so force alone will not cure them (FACT). FD-VLA claims a force token predicted from vision outperforms physical sensors, but only an abstract without numbers was read (FD-VLA).

Layering wins at task level, and compression remains nearly untested

Sourced. DeepMind's same-stack study is the strongest evidence for layering: flat 69.63 / 25.30 / 50.90 against best hierarchy 78.22 / 67.08 / 80.89 on short, long and reasoning tasks, and 12/15 against 3/15 on real ALOHA, with a closed low-level policy (Hi-VLA). TAPT raised real FR3 success from 16.7% to 58.3% with bounded VLA tools at 30 demos per task and uses a progress signal to request replanning (TAPT). The hazard is narrow fine-tuning, which dropped π0.5 from 73.3 to 32.0 inside the hierarchy (Hi-VLA). On the other side, DreamZero, a flat 14B world-action model trained only on DROID, led RoboArena in April at 1750 to π0.5's 1622 (NVIDIA). GEN-1.5 coverage still repeats company numbers with no independent evaluation found (eWeek). Robocurve's flat LLM loop needs no fast layer to reach 45 to 48%, and RAX's gains come from taking the reasoner out of the loop. TypeGo's compiled reflexes preempt within 50 ms but on one quadruped with 10 trials per task, and improved latency more than success (TypeGo). The only direct gesture at compressing deliberation into a fine-tuned model is World Action Agent, in simulation and work in progress (WAA).

Inference. Scaling is winning at skill level and hierarchy at task level, and a reflex layer that improves through scaling is compatible with both. Evidence supports asymmetric coupling, a fast layer with authority and a clean upward failure signal, more than continuous supervision. A wrist-only rig runs DROID-derived checkpoints off-spec, which removes the obvious π0.5 shortcut unless a fixed camera is added. Finding #154 puts the local VLM at 6.0 GB and cuRobo at 546 MB, which leaves about 9.5 GB, barely enough for π0.5's measured 8.87 GiB, so the VLM would need unloading. A learned reflex should be a fixed-instruction tool selected by the orchestrator, which sidesteps the steerability hazard, and SmolVLA as an LLM-called tool remains untested.

Evidence quality. Hierarchy benefit: moderate, strongest from one lab on one stack and not tested against frontier flat models. VLA independent evaluations: low to moderate, small trial counts and non-Franka arms. Memory and latency: vendor or third-party figures on faster GPUs.

MHS stays gated while LeRobot, deadlines and regulation keep moving

Sourced. The Model Hardware Standard shows no change since the 27 August preview: application-gated, no public specification, repository, tag syntax or open-source date (Anthropic, MHS site). LeRobot's v0.6.0 and v0.7.0 notes do not mention it, despite Hugging Face being a named participant (v0.7.0). PKU's OpenMHS is unofficial, created 7 November, with chained action receipts in an offline-verifiable log (OpenMHS). LeRobot reached v0.7.0 on 5 November with dataset format v3.0 (dataset code), and Franka arrives only through plugins such as an FR3 plugin built on `franky`, unverified on a Panda (third-party robots). Its RoboLab benchmark provides simulated DROID Panda tasks including stacking (RoboLab). SO-101 is the documented flagship, and two MCP servers already drive it (robot_MCP, so101-mcp).

No open competition found lets a Franka or SO-100 compete from home in a real-world track. LeHome 2027's simulation round opens 1 December and is open to all teams (LeHome); RGMC 12 has an interest form only (RGMC). RSS 2027's Stage 1 abstract is due 4 December, tolerates insightful negative results, and is double-blind with no GitHub links (RSS). The Agentic Physical AI workshop at ICLR 2027 is still a proposal, with a planned 1 February deadline and a rule that short papers be primarily human-authored (workshop). Agentic Robotics at CVPR 2027 is also proposed, for 12 March (workshop), and an ICRA 2027 lifelong-memory workshop accepts 1 to 2 page deployment reports, deadline to be announced (workshop). No benchmark specifies block stacking at a stated clearance; YCB and NIST templates are the best base (NIST).

EU Machinery Regulation 2023/1230 applies from 20 January 2027 after a corrigendum, unchanged by the 2026 Digital Omnibus (Commission, Omnibus). It defines supply in the course of commercial activity, even free of charge, as making available, and includes own-use builders as manufacturers (Regulation).

Inference, not legal advice. "MCP server shaped to sit above MHS" is defensible today, and "built on MHS" is not. Hebbot's public notebook makes RSS anonymity awkward, and under four weeks is too short given no promoted skill, so the ICLR short-paper track or the ICRA deployment format fits better. Non-commercial MIT code does not obviously count as making a product available, but selling kits in the EU would change that, and Doug's location was not in the materials. Substantial modification was not analysed.

Evidence quality. Primary pages for dates and legal text, high. GitHub absence checks miss private branches. Community projects are unproven.

Conclusion: finish touch stacking first, then make the learned test earn its place

The decision is to finish touch stacking, under a stop rule, and to treat its promoted form as the first reflex. The best case for testing a learned policy now is strong. Joint-torque stacking has no published precedent; a wrist-servoing policy would be implicitly calibration-free, and calibration has been Hebbot's recurring failure; and touch stacking alone does not test the distillation thesis. That case loses for now on three facts. Five successes support no neural training in any paper found, the dispatcher cannot yet run a learned body, and the defects that stopped touch stacking were bookkeeping bugs, which are cheaper to fix than a dataset is to collect. Finishing the classical path also generates that dataset and supplies the baseline that a learned reflex must beat, which the 6.6% vision figure cannot do cleanly if its attempts straddle the September bracket change. The thresholds below are my judgment calls, not published standards.

Rank	Action	Informed by	Exit test or stop rule
1	Register touch stacking as a substrate skill. Fix the raster centring, confirm the 45 mm clustering bug is gone from every code path, use `0_F_ext_z` for placement contact, add yaw and held-block probing, keep misses as evidence, re-check the #155 constants against the straightened bracket	FORCA, 2603.21410, BayesContact, notebook #135, #136, #156	Freeze code, then 30 attempts. **At 9 or fewer successes** (upper bound 49%) with lateral error as the main failure, promote rank 5 ahead of ranks 3 and 4
2	Promote the frozen composition as the first reflex. Add `Unsure`, per-run evidence, expiry on hardware change, and a pre-registered partial-credit sequential gate. Plot LLM calls, tokens, time and human interventions per verified success	RAX, ASPIRE, Kuroki, N-SCORE, SPINE	Gate passes against the deliberate path in paired trials, then 14 straight to claim 80%
3	Instrument see-versus-feel residuals with change-point detection, use them to shrink the search prior, and measure per-probe time	Notebook #155, #156; design without a published precedent	Detects a deliberately loosened bracket before grasp failure; find time well below 6.75 minutes
4	Run reset-free pick-and-place with physical labels plus a pre/post VLM	Tether, AutoDex, No Free Checker	About 50 verified successes on one

Rank	Action	Informed by	Exit test or stop rule
	check, with false positives and negatives hand-audited		frozen hardware configuration
5	Implement the `learned_policy` body. Train ACT locally on the last-centimetre segment only, or a residual on the classical reference, as a fixed-instruction tool	RCP, FILIC, FR3 fusion, PhAIL, Hi-VLA steerability, TAPT	Beats rank 2's reflex in the same paired gate; otherwise it stays unpromoted and the negative result is published
6	Add a fixed camera if one is on the shelf, then export LeRobot v3.0 datasets, apply for MHS access, send the RGMC interest form, and target ICLR or ICRA workshop formats	Camera requirements, LeRobot, venues above	Cheap; do opportunistically

Local π0.5 fine-tuning, force-aware VLAs, large world models, FR3-specific plugin work and LeHome should wait, since each buys less than ranks 1 to 5 at this data scale. The project's most distinctive available contribution is a measured curve of deliberation cost against practice across a real promotion event, which this literature review did not find anywhere on hardware. Three results would change this ranking: a published joint-torque stacking method, an independent wrist-only policy that stacks reliably from tens of demos, or Hebbot's own stop rule firing.